

Technical Specification: Text-Based Football Career Simulator Prototype (Super FAFI LegendMaker)

1. Executive Summary & Tech Stack

The goal of this project is to build a lightweight, performant, text-based football career simulation web application. The core loop consists of creating a custom player, progressing year-by-year through career choices, simulating seasonal stats using Elo-weighted mathematical models, receiving realistic transfer offers, and viewing a final career summary.

Stack

- **Frontend:** React 18+ with TypeScript, Vite, Tailwind CSS, Lucide React (Icons).
- **Backend & Storage:** Supabase (PostgreSQL) for storing pre-populated leagues and teams data.
- **State Management:** React Context / Zustand for handling active career runtime state locally (to ensure instant response times without network latency during season progression).

2. Database Schema (Supabase / PostgreSQL)

Create the following tables in Supabase. These tables serve as the static global reference data for clubs and leagues worldwide.

SQL

-- 1. LEAGUES TABLE

```
CREATE TABLE leagues (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name VARCHAR NOT NULL,  
  country VARCHAR NOT NULL,  
  confederation VARCHAR NOT NULL, -- 'UEFA', 'CONMEBOL', 'CONCACAF', 'AFC', 'CAF', 'OFC'  
  tier INT NOT NULL CHECK (tier BETWEEN 1 AND 5),  
  logo_url TEXT  
);
```

-- 2. TEAMS TABLE

```
CREATE TABLE teams (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name VARCHAR NOT NULL,  
  country VARCHAR NOT NULL,  
  confederation VARCHAR NOT NULL, -- 'UEFA', 'CONMEBOL', 'CONCACAF', 'AFC', 'CAF', 'OFC'  
  tier INT NOT NULL CHECK (tier BETWEEN 1 AND 5),  
  logo_url TEXT  
);
```

```

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
league_id UUID REFERENCES leagues(id) ON DELETE CASCADE,
name VARCHAR NOT NULL,
elo_rating INT NOT NULL CHECK (elo_rating BETWEEN 30 AND 99),
reputation INT NOT NULL CHECK (reputation BETWEEN 1 AND 100),
logo_url TEXT
);

-- INDEXES FOR FAST QUERYING
CREATE INDEX idx_teams_league ON teams(league_id);
CREATE INDEX idx_teams_elo ON teams(elo_rating);
CREATE INDEX idx_leagues_tier_confed ON leagues(tier, confederation);

```

3. Mathematical & Simulation Engine

3.1 Player Valuation & Budget Gate

Market Value (V) is determined by player Rating (R) and Age (A):

$$V = \text{BaseValue}(R) * \text{AgeMultiplier}(A)$$

- **Base Value:**

$$\text{BaseValue}(R) = 10^{((R - 40) / 12)} * 100,000$$
- **Age Multiplier (A):**
 - Age 16–21: 1.3x - 1.5x (High potential premium)
 - Age 22–28: 1.0x (Prime)
 - Age 29–33: 0.6x - 0.3x (Depreciating)
 - Age 34+: 0.1x - 0.2x (Minimal transfer value)
- **Team Purchasing Power Gate (B_max):**
 A team with Elo rating E_{team} can spend a maximum of:

$$B_{\text{max}} = 100,000 * (E_{\text{team}} / 35)^{3.5}$$

Rule: If $V > B_{\text{max}}$, the team **cannot** generate a transfer offer for the player.

3.2 Seasonal Match & Stats Engine

Calculates expected Goals and Assists based on player rating vs. average league Elo (E_{league}):

$$\text{Performance Ratio (P)} = R_{\text{player}} / E_{\text{league}}$$

$$\text{Expected Output} = \text{BasePosOutput} * P^2 + \text{RNGVariance}(-3, +3)$$

- **Base Positions Outputs:**
 - **ST/CF:** ~15 Goals / 4 Assists
 - **Winger/CAM:** ~9 Goals / 9 Assists

- **CM/CDM:** ~3 Goals / 6 Assists
- **Defender:** ~1 Goal / 2 Assists

3.3 Transfer Probability Matrix

When generating yearly transfer offers, evaluate candidate teams using this probability weight formula:

Weight = $W_{\text{Tier}} * W_{\text{Region}} * W_{\text{Nationality}}$

Weight Factor	Condition	Multiplier
W Tier	$ \text{Tier_League} - \text{Tier_Ideal} = 0$	\$1.0\$
	Step up (+1 Tier) or Step down (-1 Tier)	\$0.35\$
	Step down (-2+ Tiers)	\$0.05\$
W Region	Same Confederation	\$1.0\$
	Historical Cross-Region (e.g., CONMEBOL → UEFA)	\$0.5\$
	Unrelated Confederation	\$0.05\$
W Nationality	Team from Player's Home Nation	\$3.0\times\$

4. Frontend Application Architecture & UI Specs

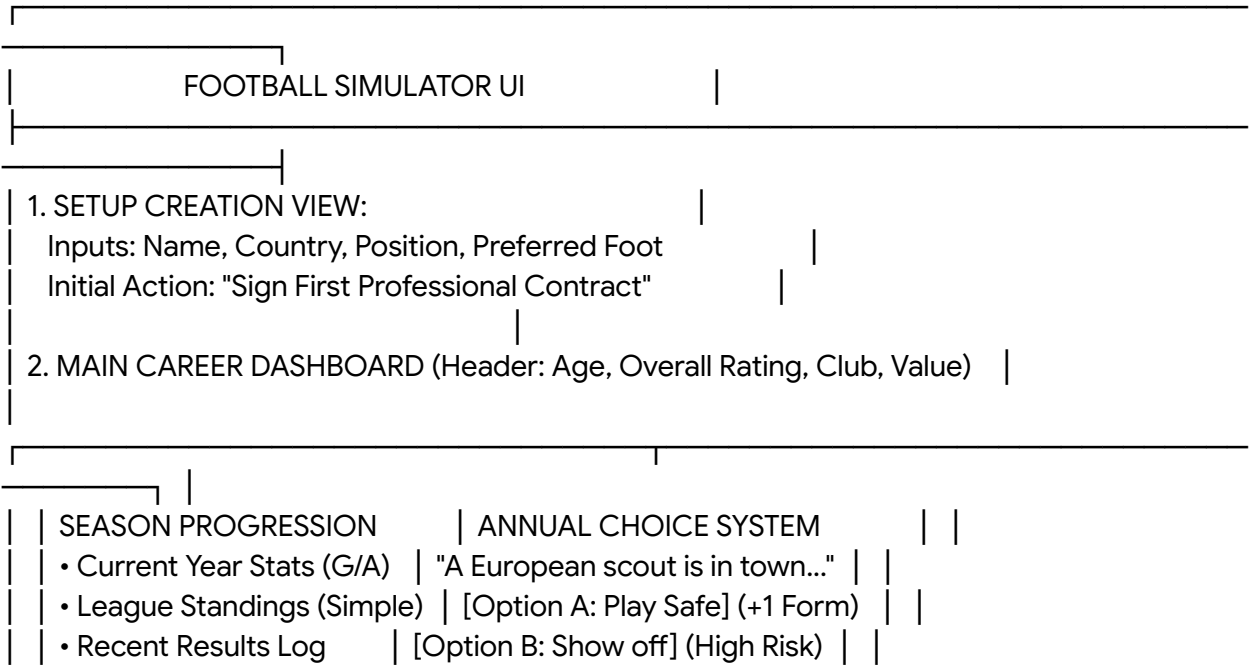
4.1 State Management (useCareerStore / Context)

```
TypeScript
export interface PlayerState {
  name: string;
```

```
nationality: string;
position: 'ST' | 'LW' | 'RW' | 'CAM' | 'CM' | 'CDM' | 'CB' | 'LB' | 'RB';
dominantFoot: 'Left' | 'Right';
age: number;
overallRating: number;
currentTeam: Team;
statsHistory: SeasonStats[];
trophies: string[];
isRetired: boolean;
}
```

```
export interface SeasonStats {
  year: number;
  teamName: string;
  appearances: number;
  goals: number;
  assists: number;
  averageRating: number;
  trophiesWon: string[];
}
```

4.2 Views & UI Layout Specification



- | |
|---|
| 3. TRANSFER WINDOW MODAL (End of Season): |
| List of generated offers from eligible teams with salary & rating |
| [Stay at Current Club] vs [Accept Offer from Team X] |
| 4. CAREER SUMMARY VIEW (Post-Retirement): |
| Total Apps, Total G/A, Trophy Cabinet, Career Rating Grade |

5. Prototype Implementation Plan for AI Agent

Instructions for the AI Coding Agent:

- 1. Setup Supabase Schema:**
 - Generate migration files or script standard SQL to create leagues and teams tables.
 - Provide a seed script (seed.json / seed.sql) containing at least 20 sample teams spanning Tier 1 to Tier 4 leagues across UEFA, CONMEBOL, and AFC.
- 2. Build Math Utility Functions (/lib/engine.ts):**
 - Implement calculatePlayerValue(rating, age)
 - Implement calculateMaxBudget(elo)
 - Implement simulateSeasonStats(playerPosition, playerRating, teamElo, leagueElo)
 - Implement generateTransferOffers(player, allTeams, allLeagues)
- 3. Build React Interfaces (/components):**
 - PlayerCreation.tsx
 - CareerDashboard.tsx
 - ChoiceModal.tsx
 - TransferOffersModal.tsx
 - CareerSummary.tsx
- 4. Game Loop Workflow:**
 - User creates player → Given choices between 4 different teams. High probability teams shown are local tier 3/4 teams, small chance of getting a local tier 1/2 team (if available), very small chance of getting a tier 2+ foreign team. If local league is too small, offer choices among other similar leagues in the same confederation.
 - "Simulate Season" triggers seasonal decision event → calculates G/A → updates stats.
 - Triggers Transfer Window if age < 38.
 - If age ≥ 34\$, prompt optional retirement, forcing retirement at age 40 (goalkeepers can play until 44).